ANSWER SCHEME
SESSION 2017/2018

NO	ANSWER SCHEME	MARKS
1	$3^{2x+1}-(16)3^x+5=0$	
	$3\left(3^x\right)^2-16\left(3^x\right)+5=0$	K1 – 3^x
	Let $m=3^x$	J1
	$3m^2-16m+5=0$	
	$(3m-1)(m-5)=0$	
	$m=\frac{1}{3}\qquad m=5$	J1
	$3^x=3^{-1}\quad 3^x=5$	K1
	$x=-1\qquad x=1.465$	J1 J1
	TOTAL	6 marks
2	$3\log_9 x=\left(\log_3 x\right)^2$	
	$\frac{3\log_3 x}{\log_3 9}=\left(\log_3 x\right)^2$	K1 – same base
	$\frac{3\log_3 x}{\log_3 3^2}=\left(\log_3 x\right)^2$	K1 – change $9=3^2$
	$\frac{3\log_3 x}{2}=\left(\log_3 x\right)^2$	K1 – log rule
	let $u=\log_3 x$	J1
	$\frac{3u}{2}=(u)^2$	
	$2u^2-3u=0$	K1 - quadratic equation and equate with zero
	$u(2u-3)=0$	
	$u=0\qquad u=\frac{3}{2}$	J1 – both
	$\log_3 x=0\qquad \log_3 x=\frac{3}{2}$	K1 – either one to find x
	$x=1\qquad x=\sqrt{27}$	J1
	$x=5.196$	J1
	TOTAL	9 marks

NO	ANSWER SCHEME	MARKS
3(a)	$z = 2 + i$	
	$\bar{z} - \frac{1}{\bar{z}} = 2 - i - \frac{1}{2-i}$	B1
	$= 2 - i - \left(\frac{1}{2-i} \times \frac{2+i}{2+i}\right)$	K1 – multiply conjugate
	$= 2 - i - \left(\frac{2+i}{5}\right)$	
	$= \frac{5(2-i) - 2 - i}{5}$	K1 – equalize denominator
	$= \frac{10 - 5i - 2 - i}{5}$	
	$= \frac{8 - 6i}{5}$	
	$= \frac{8}{5} - \frac{6}{5}i$	J1
3(b)	$\left\lvert \bar{z} - \frac{1}{\bar{z}} \right\rvert = \left\lvert \frac{8}{5} - \frac{6}{5}i \right\rvert$	
	$= \sqrt{\left(\frac{8}{5}\right)^2 + \left(-\frac{6}{5}\right)^2}$	K1 - modulus
	$= 2$	J1
	$\left(\bar{z} - \frac{1}{\bar{z}}\right)^2 = \left(\frac{8}{5} - \frac{6}{5}i\right)\left(\frac{8}{5} - \frac{6}{5}i\right)$	
	$= \frac{64}{25} - 2\left(\frac{48}{25}\right)i - \frac{36}{25}$	K1 - expansion
	$= \frac{28}{25} - \frac{96}{25}i$	J1
	$\alpha + \beta i = \left\lvert \bar{z} - \frac{1}{\bar{z}} \right\rvert \left(\bar{z} - \frac{1}{\bar{z}}\right)^2$	
	$= 2\left(\frac{28}{25} - \frac{96}{25}i\right)$	K1 – multiply 2
	$= \frac{56}{25} - \frac{192}{25}i$	
	compare	K1 – compare
	$\alpha = \frac{56}{25}$ $\quad \beta = -\frac{192}{25}$	J1 J1
	TOTAL	**12 marks**

NO	ANSWER SCHEME	MARKS
4(a)	$\frac{5}{x+3}-1\le 0$ $\frac{5-x-3}{x+3}\le 0$ $\frac{2-x}{x+3}\le 0$ C.N: $x=-3, x=2$ -3 2 *Interval* $(-\infty,-3)$ $(-3,2]$ $[2,\infty)$ *Value x* -5 0 3 $2-x$ $-$ $-$ $+$ $x+3$ $-$ $+$ $+$ Conc $\boxed{+}$ - $\boxed{+}$ $(-\infty,-3)\cup[2,\infty)$	K1 – equate denominator K1 – simplify K1 J1
4(b)	$\left\lvert\frac{3x-2}{2x+3}\right\rvert > 2$ $\frac{3x-2}{2x+3} > 2$ or $\frac{3x-2}{2x+3} < -2$ $\frac{3x-2}{2x+3} - 2 > 0$ $\quad$ $\frac{3x-2}{2x+3} + 2 < 0$ $\frac{3x-2-2(2x+3)}{2x+3} > 0$ $\quad$ $\frac{3x-2+2(2x+3)}{2x+3} < 0$ $\frac{-x-8}{2x+3} > 0$ $\quad$ $\frac{7x+4}{2x+3} < 0$ C.N: $x=-8, x=-\frac{3}{2}$ -8 $-\frac{3}{2}$ *Interval* $(-\infty,-8)$ $\left(-8,-\frac{3}{2}\right)$ $\left(-\frac{3}{2},\infty\right)$ *Value x* -9 -2 0 $-x-8$ + – - $2x+3$ – - + Conc - $\boxed{+}$ - $\left(-8,-\frac{3}{2}\right)$	B1 - definition K1 – collect one side J1 - both K1 - method J1

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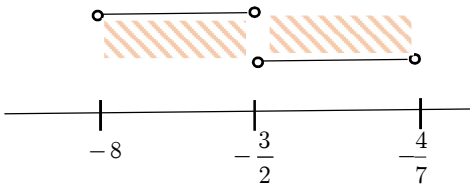
4(b)	C.N: $x = -\frac{3}{2}, x = -\frac{4}{7}$	

$-\frac{3}{2}$ $\quad$ $-\frac{4}{7}$

Interval	$\left(-\infty, -\frac{3}{2}\right)$	$\left(-\frac{3}{2}, -\frac{4}{7}\right)$	$\left(-\frac{4}{7}, \infty\right)$
Value x	-3	-1	0
$7x+4$	-	–	+
$2x+3$	–	+	+
Conc	+	$\boxed{-}$	+

K1 – method

$\left(-\frac{3}{2}, -\frac{4}{7}\right)$ — J1

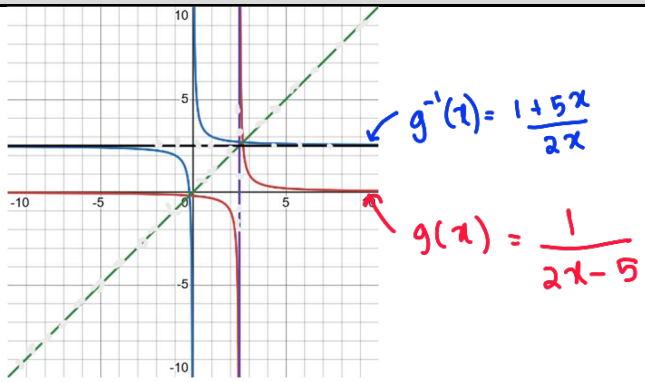


K1 - combine

$\left(-8, -\frac{3}{2}\right) \cup \left(-\frac{3}{2}, -\frac{4}{7}\right)$ — J1

TOTAL		13 marks
5(a)	$f(x) = (x-2)^2 + 1,\ x > 2$ and $g(x) = \ln(x+1),\ x > 0$	
	$f(x) = (x-2)^2 + 1$	
	$f\left[f^{-1}(x)\right] = x$	
	$\left(f^{-1}(x) - 2\right)^2 + 1 = x$	K1 – method to find inverse
	$\left(f^{-1}(x) - 2\right)^2 = x - 1$	
	$f^{-1}(x) - 2 = \pm\sqrt{x-1}$	
	$f^{-1}(x) = \pm\sqrt{x-1} + 2$	K1 – solving for inverse
	$\therefore f^{-1}(x) = \sqrt{x-1} + 2\ ,\ x > 2$	J1
	$g(x) = \ln(x+1)$	
	$g\left[g^{-1}(x)\right] = x$	
	$\ln\left(g^{-1}(x) + 1\right) = x$	K1 - method to find inverse
	$g^{-1}(x) + 1 = e^x$	
	$g^{-1}(x) = e^x - 1$	J1
	$D_{f^{-1}}:\ (1, \infty)$ $\quad$ $R_{f^{-1}}:\ (2, \infty)$	B1 B1
	$D_{g^{-1}}:\ (0, \infty)$ $\quad$ $R_{g^{-1}}:\ (0, \infty)$	B1 B1

NO	ANSWER SCHEME	MARKS
5(b)	$(g \circ f)(x) = g\left[(x-2)^2 + 1\right]$	
	$= \ln\left[\left((x-2)^2 + 1\right) + 1\right]$	K1 – subs $f(x)$
	$= \ln\left[(x-2)^2 + 2\right]$	J1
	$(g \circ f)(2) = \ln\left[(2-2)^2 + 2\right]$	K1 – subs $x = 2$
	$= \ln 2$	
	$= 0.693$	J1
	TOTAL	**13 marks**
6(a)	$g(x) = \dfrac{1}{2x-5}$	
	$2x - 5 \neq 0$	
	$x \neq \dfrac{5}{2}$	K1
	$D_g : \left(-\infty, \dfrac{5}{2}\right) \cup \left(\dfrac{5}{2}, \infty\right)$	J1
	$R_g : (-\infty, 0) \cup (0, \infty)$	J1
6(b)	$g(x_1) = g(x_2)$	
	$\dfrac{1}{2x_1 - 5} = \dfrac{1}{2x_2 - 5}$	K1 - state
	$2x_2 - 5 = 2x_1 - 5$	
	$2x_1 = 2x_2$	
	$x_1 = x_2$	
	$\therefore g(x)$ is one-to-one function.	J1
	$g\left[g^{-1}(x)\right] = x$	
	$\dfrac{1}{2\left[g^{-1}(x)\right] - 5} = x$	K1 – using properties
	$2\left[g^{-1}(x)\right] - 5 = \dfrac{1}{x}$	K1 – attempt to find inverse
	$2\left[g^{-1}(x)\right] = \dfrac{1}{x} + 5$	
	$2\left[g^{-1}(x)\right] = \dfrac{1+5x}{x}$	
	$g^{-1}(x) = \dfrac{1+5x}{2x}$	J1

NO	ANSWER SCHEME	MARKS
6(c)		R1 – shape and label R1 – with pair R1 – all correct (asymptote, intercept)
6(d)	$g \circ g^{-1}(x) = \dfrac{1}{2\left(\dfrac{1+5x}{2x}\right)-5}$ $= \dfrac{1}{\dfrac{1+5x}{x}-5}$ $= \dfrac{x}{1+5x-5x}$ $= x$	K1 – to get $g \circ g^{-1}$ J1
	TOTAL	**13 marks**
7(a)	$\lim\limits_{x\to 2} \dfrac{x^3-8}{x^2-2x} = \lim\limits_{x\to 2} \dfrac{(x-2)(x^2+2x+4)}{x(x-2)}$ $= \lim\limits_{x\to 2} \dfrac{x^2+2x+4}{x}$ $= \dfrac{2^2+2(2)+4}{4}$ $= 6$	K1 – factorize K1 - simplify J1 - answer
7(b)	$\lim\limits_{x\to\infty} \sqrt{\dfrac{5x+7}{6x-5}} = \sqrt{\lim\limits_{x\to\infty} \dfrac{5x+7}{6x-5}}$ $= \sqrt{\lim\limits_{x\to\infty} \dfrac{\dfrac{5x}{x}+\dfrac{7}{x}}{\dfrac{6x}{x}-\dfrac{5}{x}}}$ $= \sqrt{\lim\limits_{x\to\infty} \dfrac{5+\dfrac{7}{x}}{6-\dfrac{5}{x}}}$ $= \sqrt{\dfrac{5}{6}}$	K1 K1 J1
	TOTAL	**6 marks**

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8	$y = e^{-2x}\sin 3x$ $u = e^{-2x} \qquad v = \sin 3x$ $u' = -2e^{-2x} \qquad v' = 3\cos 3x$	K1 – differentiate
	$\frac{dy}{dx} = -2e^{-2x}\sin 3x + 3e^{-2x}\cos 3x$ $= e^{-2x}\left[3\cos 3x - 2\sin 3x\right]$	J1 - $\frac{dy}{dx}$
	$u = e^{-2x} \qquad v = -2\sin 3x + 3\cos 3x$ $u' = -2e^{-2x} \qquad v' = -6\cos 3x - 9\sin 3x$	K1 – differentiate
	$\frac{d^2y}{dx^2} = (-2\sin 3x + 3\cos 3x)(-2e^{-2x}) + (-6\cos 3x - 9\sin 3x)(e^{-2x})$ $= e^{-2x}\left[(4\sin 3x - 6\cos 3x) + (-6\cos 3x - 9\sin 3x)\right]$ $= e^{-2x}\left[-5\sin 3x - 12\cos 3x\right]$	J1 - $\frac{d^2y}{dx^2}$
	$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 13y$ $= e^{-2x}\left[-5\sin 3x - 12\cos 3x\right] + 4\left(e^{-2x}\left[-2\sin 3x + 3\cos 3x\right]\right) + 13\left(e^{-2x}\sin 3x\right)$ $= e^{-2x}\left[-5\sin 3x - 12\cos 3x - 8\sin 3x + 12\cos 3x + 13\sin 3x\right]$ $= e^{-2x}(0)$	K1 – substitute
	$= 0$ (shown)	J1 - answer
	TOTAL	**6 marks**
9	The condition for continuity of $f(x)$ at $x = a$ (a) $f(a)$ is defined. (b) $\lim_{x\to a} f(x)$ is exist. (c) $f(a) = \lim_{x\to a} f(x)$	K1J1
9(a)	At $x = 0$ $f(0) = \lim_{x\to 0^-}(n - 2\cos x) = \lim_{x\to 0^+}(2 + mx^2)$	B1
	$\lim_{x\to 0^-}(n - 2\cos x) = \lim_{x\to 0^+}(2 + mx^2)$	K1
	$n - 2 = 2$ $n = 4$	J1
	At $x = 2$ $f(2) = \lim_{x\to 2^-}(2 + mx^2) = \lim_{x\to 2^+}(m - x)$	B1
	$\lim_{x\to 2^-}(2 + mx^2) = \lim_{x\to 2^+}(m - x)$	K1
	$2 + m(2)^2 = m - 2$ $m = \frac{-4}{3}$	J1

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9(b)	$m=-2,\ n=4$	
	$f(x)=\begin{cases} 4-2\cos x, & x<0 \\ 2-2x^2, & 0\le x<2 \\ -2-x, & x\ge 2 \end{cases}$	
	$f'(2^-)=\lim\limits_{x\to 2^-}\dfrac{\left(2-2x^2\right)-\left(-4\right)}{x-2}$	K1
	$=\lim\limits_{x\to 2^-}\dfrac{-2x^2+6}{x-2}$	
	$=+\infty$ (undefined)	
	$f'(2^+)=\lim\limits_{x\to 2^+}\dfrac{\left(-2-x\right)-\left(-4\right)}{x-2}$	K1
	$=\lim\limits_{x\to 2^+}\dfrac{-\left(x-2\right)}{x-2}$	
	$=-1$	J1 – both
	Since $f'(2^-)\neq f'(2^+)$, therefore $f(x)$ is not differentiable at $x=2$	J1
	TOTAL	**12 marks**
10(a)	$x^2-3y^2=ae^{y-2x}+by-6$	
	At the point $(1,2)$	
	$(1)^2-3(2)^2=ae^{2-2(1)}+b(2)-6$	K1 -Substitute
	$a+2b=-5$ ... (i)	K1 – 1st equation
	$2x-6y\dfrac{dy}{dx}=ae^{y-2x}\dfrac{d}{dx}(y-2x)+b\dfrac{dy}{dx}$	
	$2x-6y\dfrac{dy}{dx}=ae^{y-2x}\left[\dfrac{dy}{dx}-2\right]+b\dfrac{dy}{dx}$	K1 - differentiate
	At the point $(1,2)$, $\dfrac{dy}{dx}=1$	
	$2(1)-\left[6(2)(1)\right]=ae^{2-2(1)}\left[1-2\right]+b(1)$	K1 - substitute
	$-a+b=-10$	
	a - $b=10$ ... (ii)	K1 – 2nd equation
	$(i)-(ii)$	K1 – any method to find a and b
	$3b=-15$	
	$b=-5$	
	$a+2(-5)=-5$	
	$a=5$	
	$\therefore a=5,\quad b=-5$	J1, J1

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10(b)	$2x-6y\frac{dy}{dx}=5e^{y-2x}\left[\frac{dy}{dx}-2\right]-5\frac{dy}{dx}$	
	$2-\left[6y\frac{d^2y}{dx^2}+6\left(\frac{dy}{dx}\right)^2\right]=\left[\left(\frac{dy}{dx}-2\right)\left(5e^{y-2x}\left(\frac{dy}{dx}-2\right)\right)+\left(5e^{y-2x}\right)\frac{d^2y}{dx^2}\right]-5\frac{d^2y}{dx^2}$	K1 - differentiate
	At the point $(1,2)$, $\frac{dy}{dx}=1$	
	$2-\left[6(2)\frac{d^2y}{dx^2}+6(1)^2\right]=\left[(1-2)\left(5e^{2-2(1)}(1-2)\right)+\left(5e^{2-2(1)}\right)\frac{d^2y}{dx^2}\right]-5\frac{d^2y}{dx^2}$	K1 – substitute
	$2-12\frac{d^2y}{dx^2}-6=5+5\frac{d^2y}{dx^2}--5\frac{d^2y}{dx^2}$	K1 - simplify
	$-12\frac{d^2y}{dx^2}=9$	
	$\frac{d^2y}{dx^2}=-\frac{3}{4}$	J1
	TOTAL	**12 marks**
11(a)	$f(x)=\frac{\ln(x-1)}{x-1}$	
	$u=\ln(x-1)$ $\quad v=x-1$	K1 - differentiate
	$u'=\frac{1}{x-1}$ $\quad v'=1$	
	$f'(x)=\frac{(x-1)\left(\frac{1}{x-1}\right)-\ln(x-1)(1)}{(x-1)^2}$	
	$=\frac{1-\ln(x-1)}{(x-1)^2}$	K1 – 1st derivative
	$f'(x)=0$ $\quad f'(x)$ undefined	
	$\frac{1-\ln(x-1)}{(x-1)^2}=0$	K1 – equate with zero
	$\ln(x-1)=1$ $\quad x-1=0$	
	$x-1=e$ $\quad x=1$ (rejected because $x>1$)	
	$x=e+1$	J1 – value of x

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11(a)	$u = 1 - \ln(x-1) \qquad v = (x-1)^2$ $u' = -\dfrac{1}{x-1} \qquad v' = 2(x-1)$ $f''(x) = \dfrac{(x-1)^2\left(-\dfrac{1}{x-1}\right) - \left(1-\ln(x-1)\right)\left(2(x-1)\right)}{(x-1)^4}$ $= \dfrac{-(x-1) - \left(1-\ln(x-1)\right)\left(2(x-1)\right)}{(x-1)^4}$ When $x = e+1$ $f''(e+1) = \dfrac{-(e+1-1) - \left(1-\ln(e+1-1)\right)\left(2(e+1-1)\right)}{(e+1-1)^4}$ $= \dfrac{-e-(1+\ln 1)(2e)}{e^4}$ $= \dfrac{-3e}{e^4}$ $= \dfrac{-3}{e^3} < 0 \quad (\text{maximum})$	K1 - differentiate K1 – 2nd derivative K1 - substitute J1
11(b)	When $x = e+1$ $f(e+1) = \dfrac{\ln(e+1-1)}{e+1-1}$ $= \dfrac{\ln e}{e}$ $= \dfrac{1}{e}$ So the exact coordinates of the maximum point is $\left(e+1, \dfrac{1}{e}\right)$	K1 – substitute into $f(x)$ J1
11(c)	$\dfrac{d^2y}{dx^2} = 0$ $\dfrac{-(x-1) - \left(1-\ln(x-1)\right)\left(2(x-1)\right)}{(x-1)^4} = 0$ $-(x-1) - \left(1-\ln(x-1)\right)\left(2(x-1)\right) = 0$ $-(x-1)\left[1 + 2\left(1-\ln(x-1)\right)\right] = 0$ $-(x-1)\left[3 - 2\ln(x-1)\right] = 0$	K1

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11(c)	$3-2\ln(x-1)=0$ $2\ln(x-1)=3$ $-(x-1)=0$ $x-1=0$ $x=1$ $\ln(x-1)=\frac{3}{2}$ $x-1=e^{\frac{3}{2}}$ $x=e^{\frac{3}{2}}+1$ Since $x\neq 1$, therefore $x=e^{\frac{3}{2}}+1$.	J1
	TOTAL	**12 marks**
12(a)	$x=3t-\frac{1}{t}$ $\quad y=t+\frac{3}{t}$ $\frac{dx}{dt}=3+\frac{1}{t^2}$ $\quad \frac{dy}{dt}=1-\frac{3}{t^2}$ $=\frac{3t^2+1}{t^2}$ $\quad =\frac{t^2-3}{t^2}$	K1 – differentiate
	$\frac{dy}{dx}=\frac{t^2-3}{t^2}\times\frac{t^2}{3t^2+1}$	K1 – chain rule
	$=\frac{t^2-3}{3t^2+1}$ (shown)	J1
	$u=t^2-3$ $\quad v=3t^2+1$ $u'=2t$ $\quad v'=6t$	K1
	$\frac{d}{dt}\left(\frac{t^2-3}{3t^2+1}\right)=\frac{(3t^2+1)(2t)-(t^2-3)(6t)}{(3t^2+1)^2}$ $=\frac{6t^3+2t-6t^3+18t}{(3t^2+1)^2}$	K1 – differentiate $\frac{dy}{dx}$
	$=\frac{20t}{(3t^2+1)^2}$	J1

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12(a)	$\frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{t^2-3}{3t^2+1}\right) \times \frac{t^2}{3t^2+1}$	
	$= \frac{20t}{\left(3t^2+1\right)^2} \times \frac{t^2}{3t^2+1}$	K1 – 2nd derivative
	$= \frac{20t^3}{\left(3t^2+1\right)^3}$	J1
12(b)	$\frac{dy}{dx} = \frac{t^2-3}{3t^2+1}$	
	$= \frac{1}{3} + \frac{-\frac{10}{3}}{3t^2+1}$	K1 – using long division
	$= \frac{1}{3} + \frac{-10}{3\left(3t^2+1\right)}$	J1
	$t^2 > 0$	K1
	$3t^2 > 0$	
	$3t^2 + 1 > 0 + 1$	
	$3t^2 + 1 > 1$	
	$3\left(3t^2+1\right) > 3$	
	$3 < 3\left(3t^2+1\right) < \infty$	
	$0 < \frac{1}{3\left(3t^2+1\right)} < \frac{1}{3}$	K1
	$0 < \frac{10}{3\left(3t^2+1\right)} < \frac{10}{3}$	
	$-\frac{10}{3} < -\frac{10}{3\left(3t^2+1\right)} < 0$	K1
	$\frac{1}{3} - \frac{10}{3} < \frac{1}{3} - \frac{10}{3\left(3t^2+1\right)} < \frac{1}{3} - 0$	
	$-3 < \frac{1}{3} - \frac{10}{3\left(3t^2+1\right)} < \frac{1}{3}$	K1
	$-3 < \frac{dy}{dx} < \frac{1}{3}$	J1
TOTAL		**15 marks**